

Simple Queue Service (SQS)

SQS is a **message queue** where messages are stored until processed.

Decouples sender & receiver

Ensures no data loss

FIFO ordering available

Asynchronous message processing

1. Create SQS Queue

The screenshot shows the AWS Management Console interface for creating a new SQS queue. The breadcrumb navigation at the top indicates the path: Amazon SQS > Queues > Create queue. The page title is 'Create queue'. Under the 'Details' section, the 'Type' is set to 'Standard' (selected with a radio button). Below this, a note states: 'You can't change the queue type after you create a queue.' The 'Name' field is filled with 'demo'. Under the 'Configuration' section, the 'Visibility timeout' is set to 30 seconds, and the 'Message retention period' is set to 4 days.

Visibility Timeout: When a Lambda function reads a message from SQS, that message becomes **hidden** from other consumers for 30 seconds.

Message Retention Period: If a message is **never processed**, SQS will store that message for **4 days** before deleting it automatically.

Range allowed: **1 minute** → **14 days**

Delivery Delay:

What you selected: 0 seconds

New messages appear in the queue **immediately**.

Receive Message Wait Time:

What you selected: 0 seconds

When a consumer (like Lambda or EC2) polls the queue, it returns immediately even if **no messages are available**.

Maximum Message Size:

What you selected: 1024 KiB (1 MB)

The largest allowed message size is **1 MB** (SQS maximum limit).

The screenshot shows the 'Create queue' configuration page in the AWS Management Console. The 'Configuration' section is active, showing settings for the new queue. The 'Visibility timeout' is set to 30 seconds. The 'Message retention period' is set to 4 days. The 'Delivery delay' is set to 0 seconds. The 'Receive message wait time' is set to 0 seconds. The 'Maximum message size' is set to 1024 KiB. The 'Encryption' section shows 'Server-side encryption' is enabled.

Configuration [Info](#)
Set the maximum message size, visibility to other consumers, and message retention.

Visibility timeout [Info](#)
30 Seconds
Should be between 0 seconds and 12 hours.

Message retention period [Info](#)
4 Days
Should be between 1 minute and 14 days.

Delivery delay [Info](#)
0 Seconds
Should be between 0 seconds and 15 minutes.

Receive message wait time [Info](#)
0 Seconds
Should be between 0 and 20 seconds.

Maximum message size [Info](#)
1024 KiB
Should be between 1 KiB and 1024 KiB.

Encryption [Info](#)
Amazon SQS provides in-transit encryption by default. To add at-rest encryption to your queue, enable server-side encryption.

Server-side encryption
☐ Disabled
☒ Enabled

Encryption key type

2. Create EC2 Instance

The screenshot shows the 'Instances' page in the AWS Management Console. A table lists the instances, with one instance named 'demo-instance' in the 'Running' state. The instance type is 't3.micro' and it is located in the 'eu-north-1b' availability zone.

Instances (1) [Info](#)
Last updated less than a minute ago

[Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive) All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
☐	demo-instance	i-040693972b89dbdb1	Running	t3.micro	Initializing	View alarms +	eu-north-1b

3. Stop Instance Manually

The screenshot shows the 'Instances' page in the AWS Management Console. The 'demo-instance' is selected, and the 'Stop instance' action is chosen from the 'Actions' dropdown menu.

Instances (1/1) [Info](#)
Last updated 3 minutes ago

[Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
☒	demo-instance	i-040693972b89dbdb1	Running	t3.micro	Initializing	View alarms +	eu-north-1b

- Stop instance
- Start instance
- Reboot instance
- Hibernate instance
- Terminate (delete) instance

4. Create IAM Role

aws [Search] [Alt+S] Account ID: 0950-7410-7161 Sakshi Jain

IAM > Roles > Create role

Step 1: Select trusted entity

Select trusted entity Info

Trusted entity type

- ☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case
Lambda

Choose a use case for the specified service.

Use case

- ☒ **Lambda**
Allows Lambda functions to call AWS services on your behalf.
- ☐ **AWSServiceRoleForLambda**
Allows Lambda to manage AWS resources on your behalf.

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IAM > Roles > Create role

Step 1: Select trusted entity

Step 2: Add permissions

Add permissions Info

Permissions policies (1/1116) Info

Choose one or more policies to attach to your new role.

Filter by Type: All types 1 match

Policy name	Type	Description
☒ AmazonEC2FullAccess	AWS managed	Provides full access to Amazon EC2 via th...

Set permissions boundary - optional

Cancel Previous Next

aws [Search] [Alt+S] Account ID: 0950-7410-7161 Sakshi Jain

IAM > Roles > Create role

Step 1: Select trusted entity

Step 2: Add permissions

Add permissions Info

Permissions policies (4/1116) Info

Choose one or more policies to attach to your new role.

Filter by Type: All types 4 matches

Policy name	Type	Description
☒ AmazonSQSFullAccess	AWS managed	Provides full access to Amazon SQS via th...
☒ AmazonSQSReadOnlyAccess	AWS managed	Provides read only access to Amazon SQS ...
☒ AWSLambdaSQSQueueExecutionRole	AWS managed	Provides receive message, delete message...
☐ SQSUnlockQueuePolicy	AWS managed	Provides access required to unlock a SQS ...

Set permissions boundary - optional

Cancel Previous Next

5. Create Lambda Function

5. Create Lambda Function

Copy Start Instance code

The screenshot shows the AWS re:Post forum interface. At the top, there's a search bar and navigation links. Below, a code snippet is displayed in a light gray box. The code is a Python lambda function that starts EC2 instances. Below the code, a text prompt says "To **start_instances**, enter the following code into the code editor:". Another code snippet follows, showing the updated lambda function code. On the right side, there's a vertical "FEEDBACK" button with a "top" link.

```
instances = ['i-12345cb6de4f78g9h', 'i-08ce9b2d7eccf6d26']
ec2 = boto3.client('ec2', region_name=region)

def lambda_handler(event, context):
    ec2.start_instances(InstanceIds=instances)
    print('started your instances: ' + str(instances))
```

To **start_instances**, enter the following code into the code editor:

```
import boto3
region = 'us-west-1'
instances = ['i-12345cb6de4f78g9h', 'i-08ce9b2d7eccf6d26']
ec2 = boto3.client('ec2', region_name=region)

def lambda_handler(event, context):
    ec2.start_instances(InstanceIds=instances)
    print('started your instances: ' + str(instances))
```

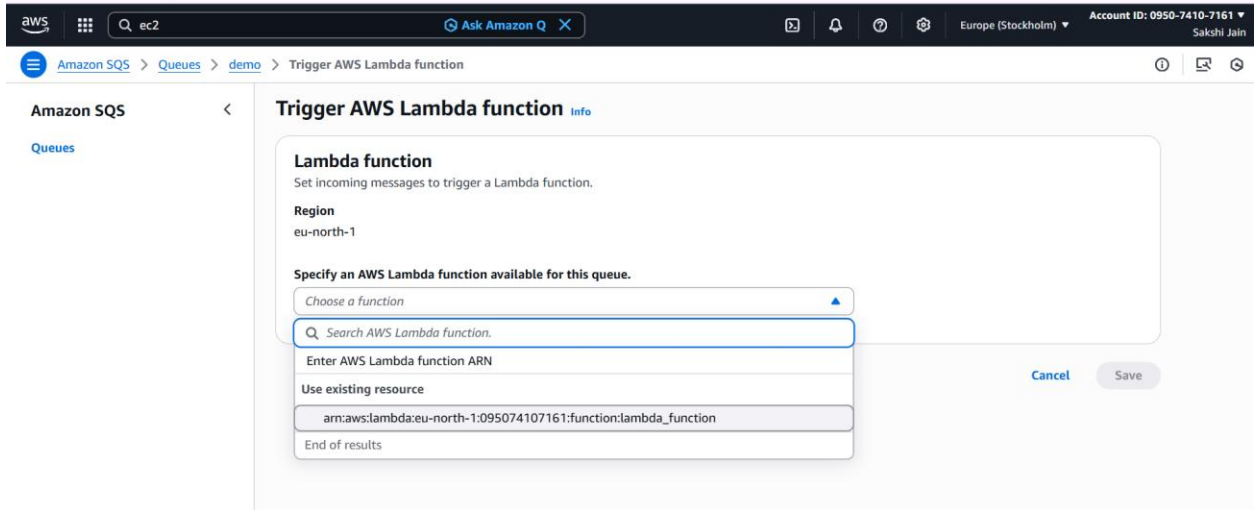
Paste code in lambda function

The screenshot shows the AWS Lambda console. The "Code source" tab is selected. The code editor displays the Python code from the previous block. The left sidebar shows the "EXPLORER" view with "LAMBDA_FUNCTION" and "lambda_function.py" listed. Below that, the "DEPLOY" section shows "Undeployed" and buttons for "Deploy (Ctrl+Shift+U)" and "Test (Ctrl+Shift+I)".

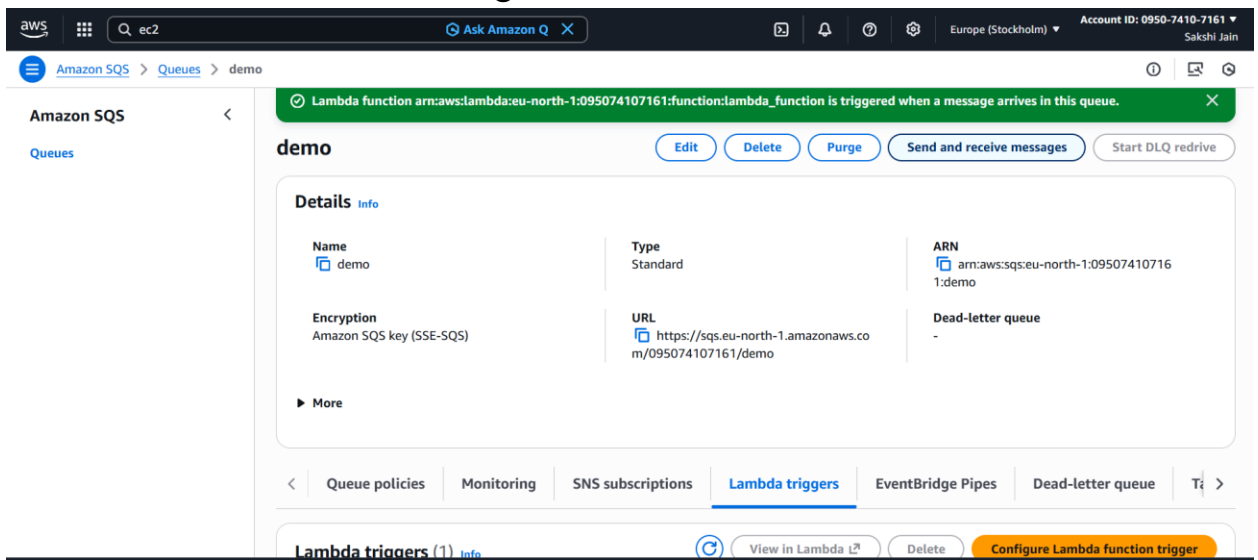
Make sure to update your – region, instance id in the code. And click on **Deploy**

This screenshot is similar to the previous one, but the code in the editor has been updated. The region is now 'eu-north-1' and the instance ID is 'i-040693972b89d8db1'. A tooltip "Amazon Q Tip 1/3: Start typing to get suggestions ([ESC] to exit)" is visible over the instance ID. The "Deploy" button is still present in the sidebar.

6. Goto SQS -> Lambda Trigger



7. Click On Send & Receive message



Write message body & click on send message

aws

ec2

Ask Amazon Q

Europe (Stockholm)

Account ID: 0950-7410-7161

Sakshi Jain

Amazon SQS

Queues

demo

Send and receive messages

Amazon SQS

Queues

Send message

Info

Message body

Enter the message to send to the queue.

Hello

Message group ID - optional, new

Info

A group identifier for the message to allow fair processing across message groups in a standard queue.

Message group ID must be 1 to 128 characters. Valid characters are a-z, A-Z, 0-9, and punctuation (!"#\$%&'()*+,-./:;<=>?@[\\]^_`{|}~).

Delivery delay

Info

The duration (in seconds) that SQS will postpone the initial delivery of the message. During this delay period, the message is not visible to consumers, allowing you to create a wait time before the message becomes available for processing.

0

Seconds

Should be between 0 seconds and 15 minutes.

Message attributes - optional

Info

Clear content

Send message

Amazon SQS

Queues

demo

Send and receive messages

Amazon SQS

Queues

Send message

Info

✓ Your message has been sent and is ready to be received.

View sent message details

✕

Message body

Enter the message to send to the queue.

Your Instance will automatically Initialized

aws

Search

[Alt+S]

Europe (Stockholm)

Account ID: 0950-7410-7161

Sakshi Jain

EC2

Instances

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instances (1/1)

Info

Last updated less than a minute ago

Connect

Instance state

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Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

< 1 >

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☒	demo-instance	i-040693972b89dbdb1	Running	t3.micro	Initializing	View alarms +	eu-north-1b